# SANKUR KUNDU

*ML Systems Engineer | LLM Training, Inference & GPU Kernels*
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**8 fixes merged in 14 days at Apple, Meta, CMU, Oxford & Hugging Face • collision-risk work under review by NASA CARA**

## TECHNICAL SKILLS

**ML systems:** PyTorch, JAX, MLX; GRPO, DDP, ZeRO/FSDP, MoE expert parallelism, paged KV cache, int8 quantization
**Languages & GPU:** Python, C++20, CUDA (tensor cores), Rust, C, Swift, SQL; Triton-style kernels, Metal, Pallas (TPU)

## OPEN-SOURCE CONTRIBUTIONS: EIGHT FIXES MERGED IN 14 DAYS, 16–29 OCT 2026

**APPLE MLX** | Apple's ML framework for Apple silicon, **29k★** | PR #4737 **Merged 22 Oct 2026**
- Fixed a data race in the backend that links Macs for distributed training (ThreadSanitizer: 2–4 races per run → **0**); reviewed, approved and merged by MLX maintainer @zcbenz.

**META EXECUTORCH** | On-device AI in Instagram & WhatsApp, **5.1k★** | #23677, #23912 **Merged 16 & 28 Oct 2026**
- **Two fixes merged** into Meta's on-device LLM kernels: closed an out-of-bounds read in the 2/4-bit embedding kernels and fixed checks aborting valid fp16-scale models; approved by Meta's Scott Roy ("LGTM! Thanks for the fix!") and Jacob Stevens.

**CMU XGRAMMAR** | Structured output in vLLM, SGLang & NVIDIA TensorRT-LLM | PR #1012 **Merged 29 Oct 2026**
- Fixed jump-forward decoding crashing whenever a grammar forces part of a UTF-8 character, a bug open since issue #132.

**HUGGING FACE OPENENV** | RL environments (Meta-PyTorch, NVIDIA, Microsoft), **2.8k★** | #1441, #1442 **Merged 19 Oct 2026**

**JAXMARL** | NeurIPS 2024; BOLD Lab: Oxford, UCL & Imperial, **868★** | PR #206 **Merged 26 Oct 2026**
- Fixed the rollouts behind 8 Q-learning baselines, which zeroed each new episode's first observation; approved and merged by lead author Alex Rutherford: "Good find!!"

**OXFORD PYZX** | Quantum-circuit optimiser, **553★** | PR #563 **Merged 22 Oct 2026**

**In review:** Google JAX (assigned to JAX co-author Matthew Johnson), vLLM (**95.7k★**), PyTorch, NVIDIA, MIT, ETH Zurich.

## HONORS & RECOGNITION

- **NASA CARA** (NASA's conjunction-assessment team): my reproduction of its Monte Carlo collision probabilities (**53/53** cases, **15–19×** faster per core) is **under review by NASA CARA**, referred by its Operations Team Lead (emails).
- **Entrepreneur First** (UK): its Talent Investor scouted me from public work alone, citing tessel's cuBLAS-parity GEMM, switchyard's MoE serving and a negative result I published, "which I rarely see" (email).

## RESEARCH (SOLE AUTHOR)

- **"Removing the Token Tax from an Existing Model"**, **"The Sampler Was Not the Policy"**, **"Reproducing NASA CARA's Monte Carlo Probability of Collision"** (arXiv, forthcoming).

## PROJECTS

**ratchet | PyTorch, C++/CUDA: RL post-training (GRPO) from scratch | study page**
- Lifted Qwen2.5-0.5B on GSM8K from **48.1% to 52.5%** in 100 steps on 2× T4, scoring all 1,319 test problems.
- Cut weight sync **369×** with in-place GPU updates, and made steps **1.9×** faster by overlapping generation with training.
- Found fp16 weight sync silently erasing most updates, so the sampler ran a policy the trainer never had (gap **5.2** → **0.04** nats).

**tessel | Rust: a tile language and compiler for GPU and TPU kernels | study page**
- Matched **cuBLAS** on GEMM (**1.00–1.04×**) and beat PyTorch SDPA on causal attention (**1.3–1.9×**) on a T4, checked against fp64.
- Built an LLM engine on these kernels alone that matches **vLLM token for token at 1.05× its throughput** (TinyLlama-1.1B).

**lipi | PyTorch: cutting the token tax on Indian languages | study page**
- Built a tokenizer that needs fewer tokens than **GPT-4o's and Gemini's in 23 of 24** Indian languages (on Hugging Face).
- Extended Qwen2.5-0.5B with it: Odia takes **8.6×** fewer tokens and, in equal GPU time, is predicted **20% better**.

**samebit | C, CUDA, PyTorch, MLX: LLM inference in integers, the same bits on every chip**
- Ran **Qwen2.5-0.5B (494M parameters) entirely in integers**: within **0.75%** of float32 perplexity and **bit-identical on an NVIDIA T4 GPU and an x86 CPU**, while float16 on the same GPU drifts by 0.25.

**More projects**
- **chirp**: GPU gravitational-wave search built up from general relativity; re-found **all 71** LIGO O4a events at **95%** of catalog SNR.
- **ink-sensitivity**: official Vesuvius ink model finds **79%** of real ink moved into an unread scroll (**AUC 0.94**).
- **relay**: fault-tolerant LLM serving on custom kernels, prefill **8.7×** faster, **384/384** requests identical through 60 kills.
- **syndrome**: GPU quantum decoder bit-identical to **IBM's**, **28.7×** faster per shot; **kessler**: matches **NASA CARA** on 53/53.

## EXPERIENCE

**GIGZEN** | Co-founder & Founding Engineer | Bhubaneswar, India **Apr 2026 – Present**
*Early-stage startup (name approved by India's MCA; incorporation in progress) | Sole engineer*
- Built Waggle, three delivery apps on one PostgreSQL backend: **53 tables, 90 RLS policies, 154 SQL functions**; **150 attacks** blocked.
- Moved dispatch, UPI reconciliation and GST billing into the database, so no client can bypass them.

## EDUCATION

**IEM KOLKATA** | B.Tech, Computer Science and Engineering (AI & ML) | Kolkata, India **Sep 2022 – Apr 2026**